Category I – Colon Interpretation

This category screens for symptoms associated with gastrointestinal dysfunction. The gastrointestinal symptoms may be due to hypochlorhydria, pancreatic insufficiency, biliary stasis, dysbiosis, yeast overgrowth, gastric or duodenal ulcer, etc. If the individual has symptoms associated with this category, a comprehensive evaluation of the gastrointestinal tract should be conducted.

Unless the gastrointestinal tract imbalance is addressed, all therapies to support imbalances at any other level will likely be unsuccessful.

- Feeling that bowels do not empty completely
- Lower abdominal pain relieved by passing stool or gas
- Alternating constipation and diarrhea
- Diarrhea
- Constipation
- Hard, dry, or small stool
- Coated tongue or "fuzzy" debris on tongue
- Pass large amount of foul-smelling gas
- More than 3 bowel movements daily
- Use laxatives frequently

Further Assessment of Category I – Colon

Further assessment is required in the following areas:

- Diet and Lifestyle
  - Check for inflammatory foods
  - Check for lack of water
  - Check for sedentary lifestyle
- Motility Issue
  - Check for brain-gut axis imbalance
  - Check for low thyroid function
- Pathogenic
  - Check for parasites—elevated eosinophils
- Autoimmune
  - Check for ASCA and/or ANCA antibodies
- Celiac
  - Check for transglutaminase antibodies
- Intestinal Permeability
  - Cyrex® Laboratories Array 2 Intestinal Antigenic Permeability Screen—occludin/zonulin antibodies, actomyosin antibodies, lipopolysaccharide antibodies

MAF Blood Chemistry Key Association Category I – Colon

- Check eosinophiles (may increase with parasitic infection)
- Check immune health
- Consider ordering a GI stool test or Cyrex® Laboratories Array 2 for intestinal permeability

MAF Nutritional Key Category I – Colon*

*Nutritional Support for the Colon:**

4-R Program: Remove, Replace, Reinoculate, and Revive

Supplements for the "Remove" phase:*

1. GI-Synergy™◊ (K64), (K83): 1 packet once a day.

Supplements for the "Replace" phase:*

1. BileMin™◊ (K11): 1-2 capsules with meals.
2. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.

Supplement for the "Reinoculate" phase:*

1. Strengtia™◊ (K61): 1-3 capsules a day.

   Alternative Formula
   - Sibiotica™◊ (K97): 1-3 capsules twice a day.

Supplements for the "Revive" phase:*

1. RepairVite™◊ (K60), (K63): 1 scoop, with up to 8 ounces of water, 1-2 times a day.

   Alternative Formulas
   - RepairVite-SE™◊ (K98): 1 scoop, with up to 4-6 ounces of water, twice a day.
   - RepairVite-GT™◊ Plus (K120): 1 scoop, with up to 8 ounces of water, once a day.
2. Strengtia™◊ (K61): 1-3 capsules a day.

   Alternative Formula
   - Sibiotica™◊ (K97): 1-3 capsules twice a day.

Please consider the Mucosal Health & Food Tolerance Program™ (from the corresponding guide)

Note: Consider pharmaceutical management to eradicate infections. The 4-R Program—which stands for remove, replace, reinoculate, and revive—should be considered to support gastrointestinal health.*

"Remove" includes supporting the microbial environment.* Individuals should work with their primary care physician to address the need to remove pathogenic organisms; however, one can use nutrients and herbs to address the terrain and support an environment in which it is less favorable for the development of bacteria, etc.* GI-Synergy™◊ (K64) combines three formulas that are each designed to support the intestinal terrain.* H-PLR™◊ (K32),

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For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Parastonil™◊ (K59), and Yeastonil™◊ (K58) are combined into one easy-to-use package for the individual.

"Replace" includes supporting the digestive system with enzymes and hydrochloric acid (EnzymixPro™◊ [K99]) and biliary support (BileMin™◊ [K11]).* EnzymixPro™◊ (K99) is a source of hydrochloric acid and enzymes that support digestive processes.* BileMin™◊ (K11) is a source of beet concentrate, taurine, vitamin C, dandelion root, milk thistle extract, ginger, and lecithin used to support the biliary system.*

To "Reinoculate" involves replenishing the gastrointestinal tract with friendly bacteria, such as *Lactobacillus and Saccharomyces boulardii.* Strengtia™◊ (K61) is a formula that offers targeted probiotics to help support the intestines.* It includes prebiotic support for enhanced potency of the probiotics. The ingredients support the bacterial environment and the intestinal immune system.*

"Revive" includes nutritional support for the gastrointestinal lining.* When reviving the gastrointestinal tract, nutrients such as L-glutamine, gamma oryzanol, aloe leaf extract, flavonoids, and deglycyrrhizinated licorice are known to support the intestinal wall structure and function.* RepairVite™◊ (K60), (K63) supports the maintenance of the intestinal tract and intestinal lining.*

At times, an individual may need a fiber supplement as a bulking agent to support bowel movements.* Fibromin™◊ (K25) offers a variety of soluble and insoluble fibers, including guar gum, pectins, lignans, cellulose, and enzymes.

Category II – Intestinal Barrier Interpretation

This category screens for symptoms associated with intestinal permeability, also known as "leaky gut." The functional laboratory markers for intestinal permeability of abnormal (high or low) functional changes of total protein and globulin are identical to hypochlorhydria, so symptom evaluation may help differentiate the appropriate mechanisms. Intestinal permeability can be appropriately identified by the Cyrex® Laboratories Array 2. The profile will measure antibodies for lipopolysaccharides, actomyosin, and occludin/zonulin. If any of those markers are positive, intestinal permeability is objectively confirmed.

If intestinal permeability is confirmed, then the mechanism for its development must be identified.

Common causes for intestinal permeability are undiagnosed gluten intolerance or celiac disease. Intestinal permeability can be evaluated by Cyrex® Laboratories Array 3 serum laboratory test. The use of medications and antibiotics, as well as alcohol abuse, systemic inflammation, and elevated stress responses may also be causative factors in intestinal permeability.

- Increasing frequency of food reactions
- Unpredictable food reactions
- Aches, pains, and swelling throughout the body
- Unpredictable abdominal swelling
- Frequent bloating and distention after eating

Further Assessment of Category II – Intestinal Barrier

Further assessment is required in the following areas:

- Check for intestinal permeability with Cyrex® Laboratories Array 2 for evaluation of:
  - Occludin/zonulin antibodies
  - Actomyosin antibodies
  - Lipopolysaccharide antibodies
- Identify mechanism of intestinal permeability

MAF Blood Chemistry Key Association Category II – Intestinal Barrier

Protein and globulin increased or decreased.

MAF Nutritional Key Category II – Intestinal Barrier*

*Nutritional Support for the Intestinal Barrier:**

1. RepairVite™◊ (K60), (K63): 1 scoop, with up to 8 ounces of water, 1-2 times a day.

   Alternative Formulas
   - RepairVite-SE™◊ (K98): 1 scoop, with up to 4-6 ounces of water, once a day.
   - RepairVite-GT™◊ Plus (K120): 1 scoop, with up to 8 ounces of water, once a day.
2. EnteroVite™◊ (K100): 1 capsule 3 times a day.
3. Trizomal™◊ Glutathione (K122), (K129A): 5 mL (approx. 1 teaspoon or 1 packet) 1-2 times a day.

Additional Support Considerations:

1. GI-Synergy™◊ (K64), (K83): 1 packet, twice a day, with a meal.
2. Strengtia™◊ (K61): 1-3 capsules a day.

   Alternative Formula
   - Sibiotica™◊ (K97): 1-3 capsules twice a day.

MAF Nutritional Key
Category II – Intestinal Barrier* (Cont.)

3. Turmero™◊ Active (K75), (K109): 5 mL (approx. 1 teaspoon) 1-2 times a day.
4. Resvero™◊ Active (K76), (K110): 5 mL (approx. 1 teaspoon) once a day.
5. AD-Pro™◊ (K119): 1 softgel, once a day, with a meal.

   Alternative Formulas
   - ADEK-Pro™◊ (K102): 1-3 softgels once a day.
   - Ultra-D 5000™◊ (K78): 1-4 teaspoons, once a day, with meals.
   - Ultra-D Complex™◊ (K35): 1-4 teaspoons, once a day, with meals.

For additional vitamin D, consider Liqua-D™◊ (K87): 1 drop once a day.

Please consider the RepairVite™◊ Program (from the corresponding guide).

Nutritional support for the gastrointestinal system includes the RepairVite™◊ (K60), (K63) powder and the associated dietary recommendations provided by the RepairVite™ Program Guide.* The use of GI-Synergy™◊ (K64) is also recommended.* Strengtia™◊ (K61), a multi-use probiotic with arabinogalactan, is also useful for nutritional support of intestinal flora.* Ultra-D Complex™◊ (K35) or Ultra-D 5000™◊ (K78) can be used as a nutritional support for the intestinal barrier and for the immune system.*

Category III – Chemical Tolerance Interpretation

This category screens for symptoms associated with a loss of chemical tolerance. A loss of chemical tolerance occurs with a breakdown of the intestinal barrier, the blood-brain barrier, or the pulmonary barrier, as well as from the disruption of regulatory T-cells. It may also occur from environmentally induced depletion of antioxidant systems. There are no laboratory patterns associated directly with a loss of chemical tolerance; however, patterns for intestinal permeability and vitamin D status should be evaluated, as discussed earlier.

- Intolerance to smells
- Intolerance to jewelry
- Intolerance to shampoo, lotion, detergents, etc.
- Multiple smell and chemical sensitivities
- Constant skin outbreaks

Further Assessment of
Category III – Chemical Tolerance

Further assessment is required in the following areas:

- Check for loss of chemical tolerance and self-tolerance
  - Check for failure of regulatory T-cells (vitamin D and glutathione)
  - Check for barrier permeability
  - Check for failure of glutathione recycling system
- Loss of chemical tolerance is usually associated with loss of self-tolerance and autoimmunity
- Cyrex® Laboratories Array 11 Chemical Immune Reactivity Screen

MAF Blood Chemistry Key Association
Category III – Chemical Tolerance

- Protein and globulin increased or decreased associated with intestinal permeability
- Depressed 25-hydroxyvitamin D levels associated with reduced regulatory T-cell support

MAF Nutritional Key
Category III – Chemical Tolerance*

*Dietary and Nutritional Support for the Immune System:**

1. Repair & Clear™ Program (basic or comprehensive): As directed by the Repair & Clear™ Program Guide.

Nutritional support for chemical tolerance includes strategies to support the glutathione recycling system, barrier system, and hepatic biotransformation function.* A comprehensive nutritional and dietary program is listed in the Repair & Clear™ Program Guide.

Category IV – Stomach Interpretation

(Also check for Hypochlorhydria)

Hypochlorhydria is a condition in which there is an inadequate production of hydrochloric acid (stomach acid). Although some free acid is present in gastric juice, it is not adequate to digest protein efficiently. When hypochlorhydria exists, there is inadequate acid to sterilize stomach content, which increases the susceptibility of infections by pathogenic bacteria, yeast, and parasites. The typical symptoms of hypochlorhydria include gas and bloating after meals. Gas and bloating take place because foods are not digested properly, and therefore, undergo fermentation. Offensive

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For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

breath is common due to protein putrefaction. The sense of fullness in the stomach after meals that takes place is due to excess gas from fermentation. Hypochlorhydria is a condition that should be treated with the utmost priority because it is the first step in a chain of digestive reactions that leads to the release of bile and pancreatic enzymes. Hypochlorhydria also leads to nutritional malabsorption of essential nutrients—such as zinc, vitamin B_{12}, manganese, and calcium—which require an acidic environment for absorption. Hypochlorhydria may be caused by deficiencies of zinc and B vitamins, hypothyroidism, food sensitivities, prolonged exposure to pathogenic organisms, *H. pylori* inoculation, or a loss of parietal cell function from aging.

It is important to note that chronic hypochlorhydria is often secondary to a subclinical *H. pylori* infection. In these cases, there are no gastric ulcer symptoms, just patterns that represent decreased HCl production. The *H. pylori* organisms furrow into the gastric lining and shut down parietal cell release of HCl. Therefore, with any case of chronic hypochlorhydria, always test for *H. pylori*.

- Excessive belching, burping, or bloating
- Gas immediately following a meal
- Offensive breath
- Difficult bowel movements
- Sense of fullness during and after meals
- Difficulty digesting proteins and meats; undigested food found in stools

Further Assessment of Category IV – Stomach

Further assessment is required in the following areas:

- Check for parietal cell degeneration from aging
- Check for symptoms of failure of brain-gut axis
- Check for low thyroid function
- Check for intestinal inflammation:
  - Check for infections – *H. pylori*
  - Check for gluten/celiac disease
  - Check for intestinal permeability

MAF Blood Chemistry Key Association Category IV – Stomach

- Check for possible hypochlorhydria:
  - Protein and globulin increased or decreased
  - BUN increased or decreased
  - Phosphorus decreased
- Check for possible *H. pylori* infection, which may alter acid production:
  - WBC increased or decreased
  - Neutrophils increased

MAF Nutritional Key Category IV – Stomach

Nutritional Support for Healthy Stomach Acid Levels and the Gastrointestinal System:

1. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.

   Alternative Formulas
   - HCL Pro-Zyme™◊ (Z26): 1-2 tablets with each meal.
   - SuperDigest-Zyme™◊ (Z22): 1-2 tablets with each meal.
2. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.

*Nutritional Support for pH Balance and the Gastrointestinal System:**

1. H-PLR™◊ (K32): 1-2 capsules with each meal.
2. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.

   Alternative Formulas
   - HCL Pro-Zyme™◊ (Z26): 1-2 tablets with each meal.
   - SuperDigest-Zyme™◊ (Z22): 1-2 tablets with each meal.
3. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.

To support levels of hydrochloric acid, EnzymixPro™◊ (K99) should be considered.* EnzymixPro™◊ (K99) supplies a high level of betaine HCl and includes enzymes to help support the digestion of certain sugars, starches, fibers, proteins, and fats.*

Category V – Stomach Interpretation

(Also check for Hyperacidity)

Hyperacidity is often associated with hypochlorhydria that has progressed to the point in which the lining of the stomach has been destroyed. In cases of hypochlorhydria, proteins are not digested due to a lack of HCl and subsequently putrefy. The putrefaction of protein creates a very acidic environment in the stomach that may lead to stomach lining damage and ulcers. If the individual has both categories of hypochlorhydria and hyperacidity, always suspect that the cause of hyperacidity is second to hypochlorhydria and protein putrefaction.

* This statement has not been evaluated by the Food and Drug Administration. This product is not intended to diagnose, treat, cure, or prevent any disease.

Apex Energetics™ products are intended for nutritional use and health maintenance only and should not replace or delay the use of any medication.

- Stomach pain, burning, or aching 1-4 hours after eating
- Use of antacids
- Feel hungry an hour or two after eating
- Heartburn when lying down or bending forward
- Temporary relief by using antacids, food, milk, or carbonated beverages
- Digestive problems subside with rest and relaxation
- Heartburn due to spicy foods, chocolate, citrus, peppers, alcohol, and caffeine

Further Assessment of Category V – Stomach

Further assessment is required in the following areas:

- Check for *H. pylori*
- Check for high aspirin use
- Check for stress-induced permeability
- Check for gastric cancer
- Check for autoimmunity:
  - Parietal cell antibodies
  - Intrinsic factor antibodies

MAF Blood Chemistry Key Association Category V – Stomach

- Check for hyperacidity:
  - Protein and globulin increased or decreased
  - BUN increased or decreased
  - Phosphorus decreased
- Check for possible *H. pylori* infection, which may alter acid production:
  - WBC increased or decreased
  - Neutrophils increased or decreased

MAF Nutritional Key Category V – Stomach*

*Nutritional Support for pH Balance, Healthy Stomach Acid Levels, and the Gastrointestinal System:**

1. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.

   Alternative Formulas
   - HCL Pro-Zyme™◊ (Z26): 1-2 tablets with each meal.
   - SuperDigest-Zyme™◊ (Z22): 1-2 tablets with each meal.
2. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.

*Nutritional Support for Gastrointestinal Health:**

1. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.
2. H-PLR™◊ (K32): 1-2 capsules with each meal.

*Nutritional Support for pH Balance and the Gastrointestinal System:**

1. H-PLR™◊ (K32): 1-2 capsules with each meal.
2. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.

   Alternative Formulas
   - HCL Pro-Zyme (Z26): 1-2 tablets with each meal.
   - SuperDigest-Zyme (Z22): 1-2 tablets with each meal.
3. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.
4. Myco-Zyme™◊ (Z13): 1-2 capsules with meals.

*Nutritional Support for pH Balance and the Adrenals:**

1. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.

Also Consider:

1. AdrenaCalm™◊ (KR16), (KR73): Apply in the morning, afternoon, and evening.

To support levels of hydrochloric acid and digestion, consider using EnzymixPro™◊ (K99).* Many individuals will notice supported digestion immediately when taking the enzyme they need to break down protein and prevent protein putrefaction.* Some of these individuals may, however, have increased issues with the ingestion of HCl. If an individual does not have a good response with EnzymixPro™◊ (K99), consider using Gastro-ULC™◊ (K29), which is a source of deglycyrrhizinated licorice, L-glutamine, flavonoids, bismuth, gamma oryzanol, chlorella, rhubarb, and mastic gum.* After 30 days, individuals can consider taking a small serving of HCl to support the stomach.* To support the bacterial environment in the stomach, H-PLR™◊ (K32) should be considered.* Cortisol from stress thins the lining of the stomach and makes an individual more susceptible to changes in the acidic environment.

Category VI – Pancreas Interpretation

Pancreatic insufficiency will cause symptoms similar to hypochlorhydria, but the symptoms are generally experienced after starch ingestion versus protein ingestion. Most people who have pancreatic insufficiency have it

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For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.